

Project Summary

BBFlow: Learning Conformational Ensembles of Proteins from Backbone Geometry via SE(3) Flow Matching

Overview. Protein function is governed not by a single static structure but by a *conformational ensemble* drawn from the Boltzmann distribution. Molecular dynamics (MD) simulation is the established route to such ensembles, but the cost of overcoming free-energy barriers and reaching ergodicity makes broad sampling prohibitive for most proteins. Recent deep generative MD emulators (AlphaFlow, ESMFlow, ConfDiff, BioEmu) approach this problem by fine-tuning large pre-trained folding models such as AlphaFold 2 or ESMFold, which inherit a costly dependence on Multiple Sequence Alignments (MSAs) or protein language model embeddings – so-called *evolutionary information*. This dependence is slow at inference, biases predictions toward natural sequences, and breaks down on *de novo* designs and intrinsically novel sequences where evolutionary signal is absent. The proposed work introduces **BBFlow** [1], an SE(3) flow-matching model that generates protein backbone ensembles conditioned solely on the equilibrium backbone *geometry* of the target protein, eliminating the dependence on folding-model weights and evolutionary information altogether.

Intellectual Merit. Three specific aims will be pursued. (1) *Reformulate ensemble generation as structure-conditioned flow matching*: instead of $p(x)$ conditioned on sequence, learn $p(x | x_{\text{eq}})$ on the Riemannian manifold $\text{SE}(3)^N$, with conditioning provided by the equilibrium backbone via a binned pairwise-distance map and equivariant local-frame direction vectors. (2) *Introduce a structure-conditioned prior* $p_0(x_0 | x_{\text{eq}})$ through partial geodesic interpolation between unconditional noise and x_{eq} – an idea inspired by stochastic interpolants and partial denoising, but novel in the flow-matching ensemble setting. (3) *Validate on the ATLAS MD benchmark, on a curated set of de novo proteins, and on multi-chain complexes*, comparing against AlphaFlow, ESMFlow-T, ConfDiff, BioEmu, and distilled variants of AlphaFlow on RMSF, pairwise RMSD, PCA $\mathcal{W}_2$, DCCM, transient-contact Jaccard, and per-conformation inference time. Preliminary published data already establish feasibility: BBFlow trained from scratch in three GPU-days reaches accuracy competitive with AlphaFlow-T while being roughly $40\times$ faster, generalizes to *de novo* proteins where MSA-dependent baselines collapse, and – although trained only on monomers – transfers to multi-chain assemblies. The conditional prior, the distance encoding, and the index-free architecture are each shown to be necessary in ablation.

Broader Impacts. Decoupling MD emulation from large folding models lowers the compute cost of ensemble generation by more than an order of magnitude, opening MD-quality sampling to laboratories and pipelines without TPU-scale resources. Public release of model weights, training and inference code, and benchmark scripts (<https://github.com/graeter-group/bbflow>) supports reproducibility and reuse. By extending ensemble prediction to *de novo* and multi-chain proteins – regimes uncovered by current state-of-the-art models – BBFlow enables *flexibility-aware protein design*, allowing dynamics screening to be incorporated directly into design loops for binders, biocatalysts, and allosteric switches. Trainees gain experience in geometric deep learning, Riemannian generative modelling, and quantitative biophysics, and the work provides instructional examples in NeurIPS-level ML and structural biology courses.

Keywords: flow matching; SE(3) equivariance; conformational ensemble; molecular dynamics emulation; protein backbone; de novo proteins.

Project Description

Overview, Hypothesis, and Specific Aims

Objective. The objective of the proposed work is to develop and benchmark **BBFlow**, an SE(3) flow-matching generative model that produces short-timescale, MD-consistent backbone conformational ensembles of proteins conditioned solely on the protein’s equilibrium backbone geometry, without any reliance on multiple-sequence-alignment (MSA) features, protein language model embeddings, or pre-trained folding-model weights.

Central Hypothesis. The pairwise-distance map and local-frame directional encoding of an equilibrium backbone x_{eq} contain sufficient geometric information to determine, up to MD precision, the short-timescale Boltzmann distribution $p(x \mid x_{\text{eq}})$ of backbone conformations – and conditioning a flow-matching prior on x_{eq} yields a model that is faster, more transferable, and less biased than sequence-conditioned ensemble generators built atop large folding models.

Specific Aims.

- **Aim 1 – Architecture and conditioning.** Adapt the GAFL/FrameDiff SE(3)-equivariant graph network with Clifford Frame Attention (CFA), and condition it on x_{eq} via (i) a binned C_α pairwise distance map and (ii) equivariant pairwise direction vectors expressed in residue-local frames. Remove the absolute residue-index feature to discourage memorization and enable transfer to multi-chain inputs.
- **Aim 2 – Conditional prior on SE(3)^N.** Replace the unconditional Gaussian/uniform prior of standard flow matching with a *conditional* prior $p_0(x_0 \mid x_{\text{eq}})$ constructed by partial geodesic interpolation, $x_0 = \gamma(x_{\text{uncond}}, x_{\text{eq}}, \xi)$, with $\xi \in (0, 1)$. Train end-to-end with the SE(3) flow-matching loss on the ATLAS dataset.
- **Aim 3 – Benchmarking and transferability.** Evaluate on the ATLAS test set, on a held-out set of *de novo* proteins generated by RFdiffusion and FrameFlow, and on five multi-chain protein complexes. Compare against AlphaFlow, AlphaFlow-T, ESMFlow-T, ConfDiff, BioEmu, and distilled AlphaFlow variants on six standard ensemble metrics plus per-conformation inference time, with ablations isolating the contribution of each design choice.

Significance and Background

Conformational ensembles are essential to protein function

Enzymatic catalysis [2], allosteric regulation [3], and ligand binding [4] are all governed by transitions across the free-energy landscape rather than by a single equilibrium structure. Sampling this landscape with all-atom MD [5, 6], while physically rigorous, is computationally limited: meaningful sampling for a single mid-sized protein typically demands hundreds of nanoseconds to microseconds of trajectory, a wall-clock cost that scales poorly to proteome- or design-library-scale screens.

Folding models predict structures, not ensembles

The accuracy revolution of AlphaFold 2 [7], RoseTTAFold-3T [8], and the protein language model ESMFold [9] predicts a single *equilibrium* structure per sequence and gives no direct access to conformational distributions. Their predictive power derives from *evolutionary information* encoded

either in the MSA processed by the Evoformer or implicitly in pLM weights – features that are expensive to compute and unavailable for genuinely novel (e.g. *de novo* designed) sequences.

Generative MD emulators and their limitations

A first generation of generative ensemble models adapts these folding models: AlphaFlow [10] fine-tunes AlphaFold 2 to emulate the ATLAS MD dataset [11] via flow matching; ESMFlow-T substitutes the ESMFold backbone; ConfDiff [12] replaces the structure module with a diffusion model; BioEmu [13] targets broader equilibrium ensembles via a separately trained model. All four inherit two costs from their parents: (i) per-sample inference of an Evoformer-class network, yielding $\mathcal{O}(10\text{--}30\text{s})$ per conformation, and (ii) sensitivity to MSA quality, which is structurally absent for *de novo* proteins. Empirically, AlphaFlow-T over-stabilizes ensembles (median RMSF 1.17 Å vs. MD’s 1.48 Å), AlphaFlow without templates fails on *de novo* proteins, and BioEmu does not target ATLAS-style MD emulation.

Significance of the proposed work

The proposed BBFlow model addresses each limitation in a single architectural move: by conditioning on *geometry* (x_{eq}) instead of *sequence*, it removes the Evoformer/pLM bottleneck, runs in ~ 0.8 s per conformation, and is independent of evolutionary information – the property that makes AlphaFlow brittle on *de novo* sequences. As MD itself begins from an initial structure, conditioning on x_{eq} does not impose any cost above the existing MD workflow, and a folding-model+BBFlow pipeline remains $\sim 30\times$ faster than AlphaFlow even when only a sequence is available. The work thereby reframes MD emulation as a *structure-to-ensemble* problem, with direct relevance to dynamics-aware *de novo* design [25–27].

Innovation

Geometric, evolution-free conditioning

Prior MD emulators inject conditioning via the Evoformer’s processing of an MSA or via pLM embeddings. The proposed work replaces this with two purely geometric features computed from x_{eq} :

$$s_{ij} = \text{bin}(\|z_i - z_j\|_2) \quad (22 \text{ bins, } 0\text{--}20 \text{ Å}), \quad (1)$$

$$e_{ij} = r_i^{-1} \left(\frac{z_j - z_i}{\|z_j - z_i\|_2} \right), \quad \|z_i - z_j\| < 5 \text{ Å}, \quad (2)$$

where $(r_i, z_i) \in \text{SE}(3)$ is the local frame of residue i . The first is a soft contact map – shown in MSA-based methods to be the implicit target representation of evolutionary information [9] – and the second is a strictly equivariant directional cue absent from prior ensemble generators.

A conditional prior for flow matching

Standard flow matching [14] uses an unconditional prior p_0 (Gaussian translations, uniform $\text{SO}(3)$ rotations). The proposed work introduces a *conditional* prior obtained by partial geodesic interpolation between unconditional noise and x_{eq} ,

$$x_{\text{uncond}} \sim p_{\text{uncond}}, \quad x_0 = \gamma(x_{\text{uncond}}, x_{\text{eq}}, \xi), \quad \xi = 0.2, \quad (3)$$

where $\gamma(\cdot, \cdot, 0) = x_{\text{uncond}}$ and $\gamma(\cdot, \cdot, 1) = x_{\text{eq}}$. Conceptually this is a flow-matching analogue of partial denoising in diffusion [16] and a generalization of stochastic interpolants [15] to manifold-valued conditioning. To the best of our knowledge, BBFlow is the first ensemble model on $\text{SE}(3)^N$ to adopt a structure-conditioned prior of this form.

Index-free, transferable architecture

Folding-style architectures [7, 10, 18] embed an absolute residue index, which encodes a fixed chain ordering and ties the model to single-chain inputs. The proposed architecture removes the index encoding altogether, relying only on geometric features. Ablation (Sec. 4) demonstrates that this choice both improves accuracy and – more importantly – enables zero-shot transfer to multi-chain complexes, which existing baselines cannot ingest without re-engineering.

Distinction from contemporaneous work

ESMFlow-T, AlphaFlow-T, AlphaFlow-T_{12L,dist} and Li et al. [23] attack the inference-time problem by *compressing* pre-trained folding models. The proposed work attacks it by *redefining the input representation*, eliminating the folding-model dependency rather than distilling it. MDGen [24] models full all-atom trajectories with consistent time evolution; the proposed work trades temporal consistency for a stationary distribution and gains substantial speed and robustness.

Approach

Aim 1: Geometric conditioning of an SE(3) flow

Manifold and vector field. Each backbone residue is represented by a frame $x_i = (r_i, z_i) \in \text{SE}(3)$ with $r_i \in \text{SO}(3)$ and $z_i \in \mathbb{R}^3$, and a protein of length N is an element of $\mathcal{M} \equiv \text{SE}(3)^N$. A time-dependent conditional vector field

$$v(x, t, x_{\text{eq}}) : \mathcal{M} \times [0, 1] \times \mathcal{M}_{\text{eq}} \rightarrow T_x \mathcal{M} \quad (4)$$

defines the flow ϕ_t via $\frac{d}{dt}\phi_t = v(\phi_t, t, x_{\text{eq}})$, $\phi_0 = x$. Following [17, 20], the SE(3) target vector field decomposes into rotational and translational components,

$$v_{\text{SO}(3)}(r_t, t \mid r_1) = \frac{\log_{r_t}(r_1)}{1-t}, \quad v_{\mathbb{R}^3}(z_t, t \mid z_1) = \frac{z_1 - z_t}{1-t}. \quad (5)$$

Network. The vector field is parameterized by an SE(3)-equivariant graph neural network adapted from GAFL [21], comprising 6 message-passing blocks of Clifford Frame Attention. Inputs are (i) the noised frames x_t at time t , (ii) the binned distance map s_{ij} , (iii) the local-frame direction vectors e_{ij} for residue pairs within 5 Å, and (iv) a 128-dimensional embedding of the amino-acid type. Crucially, no absolute residue index is supplied – the ordering is recovered from x_{eq} .

Aim 2: Conditional prior and training

The conditional prior $p_0(\cdot \mid x_{\text{eq}})$ is constructed by geodesic interpolation, with $\xi = 0.2$ chosen by validation. Training minimizes the SE(3) flow-matching loss

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\left\| v - \hat{v}(x_t, t, x_{\text{eq}}) \right\|_{\text{SE}(3)}^2 \right], \quad \|v\|_{\text{SE}(3)}^2 \equiv \frac{1}{2} \text{Tr}(v_r v_r^T) + \|v_z\|_2^2, \quad (6)$$

with $t \sim \mathcal{U}(0, 1)$, $(x_1, x_{\text{eq}}) \sim p_{\text{data}}$, and $x_0 \sim p_0(\cdot | x_{\text{eq}})$. An auxiliary loss following [17] stabilizes early training. We train on the ATLAS [11] split of [10] (1265/39/82 train/val/test) for 3 days on two NVIDIA A100-40GB GPUs, using 20 flow timesteps and 250 sampled conformations per protein at inference.

Aim 3: Benchmarking

Metrics. We report (i) per-residue Root Mean Square Fluctuation (RMSF) Pearson r and mean absolute error (MAE), (ii) median RMSF (probing systematic over/under-stabilization), (iii) pairwise RMSD MAE, (iv) PCA $\mathscr{W}_2$ -distance against MD on the first two principal components, (v) Pearson r of the Dynamic Cross-Correlation Matrix (DCCM), (vi) transient- contact Jaccard J_{tr} , and (vii) inference time per generated conformation on a 302-residue test protein.

Preliminary results – ATLAS benchmark. Table 1 summarises performance on the ATLAS test set. BBFlow is $\sim 40\times$ **faster** than AlphaFlow-T at comparable accuracy, achieves the best pairwise RMSD MAE (0.77 Å), and matches the MD median RMSF (1.49 Å vs. MD 1.48 Å) – avoiding the over-stabilization of AlphaFlow-T (1.17 Å).

Model	RMSF $r \uparrow$	RMSF MAE $\downarrow$	Pw-RMSD MAE $\downarrow$	DCCM $r \uparrow$	PCA $\mathscr{W}_2 \downarrow$	Time [s] $\downarrow$
AlphaFlow	0.86	0.59	1.35	0.86	1.47	32.0
ConfDiff	0.88	0.62	1.45	0.86	1.41	20.2
AlphaFlow-T	0.92	0.41	0.91	0.89	1.28	32.6
ESMFlow-T	0.92	0.52	0.94	0.89	1.48	11.2
AlphaFlow-T _{dist}	0.92	0.68	1.41	0.88	1.43	3.3
BBFlow (proposed)	0.90	<u>0.42</u>	<u>0.77</u>	0.87	<u>1.33</u>	0.8

Table 1: Performance on the ATLAS test set. BBFlow matches or improves on the distilled AlphaFlow variants while being more than an order of magnitude faster than AlphaFlow-T at comparable accuracy. Adapted from [1].

Preliminary results – de novo proteins. On 50 *de novo* proteins generated by RFdiffusion [22] and FrameFlow [19] with three 100-ns MD trajectories each, AlphaFlow without templates collapses to RMSF $r = 0.47$ and BioEmu to $r = 0.60$, whereas BBFlow reaches $r = 0.84$ with a pairwise RMSD MAE of 0.32 Å – the strongest result among methods that do not consume the equilibrium structure as a template.

Preliminary results – multi-chain proteins. On five protein dimers (1BGS, 8HXR, 5EY7, and two further complexes), DCCM correlations of 0.85, 0.72, and 0.89 are obtained between BBFlow- and MD-derived ensembles, capturing both intra- and inter-chain motions despite training only on monomers. AlphaFlow fails on the same inputs without parser modifications.

Ablation. A targeted ablation (Tab. 2) confirms that (i) the conditional prior, (ii) the direction encoding, and (iii) removal of the residue-index feature each contribute; the binned distance encoding is essential – removing it raises RMSF MAE from 0.42 to 5.88 Å. Even without amino-acid identity, BBFlow remains competitive with non-template AlphaFlow.

Variant	Cond. prior	Dist. enc.	Dir. enc.	AA enc.	Index enc.	RMSF MAE	Pw-RMSD MAE	DCCM
BBFlow (full)	✓	✓	✓	✓		0.42	0.77	<u>0.87</u>
–cond. prior		✓	✓	✓	✓	0.52	1.15	0.85
–direction enc.		✓		✓	✓	0.48	0.90	0.86
+index enc.	✓	✓	✓	✓	✓	0.42	0.82	0.88
–AA enc.	✓	✓			✓	0.54	0.93	0.85
–distance enc.	✓			✓	✓	5.88	7.08	0.55

Table 2: Ablation on the ATLAS test set. The binned distance encoding is essential; the conditional prior, direction encoding, and absence of index encoding each yield measurable improvements. Adapted from [1].

Outlook and alternative approaches

Next steps. Three follow-on directions extend the proposed work. (a) *Side-chain and all-atom ensembles*: BBFlow currently generates backbone frames only; coupling to a fixed-backbone packer or to MDGen-style atomistic generators is a natural extension for ligand-binding studies. (b) *Longer-timescale ensembles*: training on μ s-scale trajectories from D. E. Shaw or AlphaFlow-MD would expand BBFlow beyond the 300-ns regime captured by ATLAS. (c) *Integration with de novo design*: coupling BBFlow with RFdiffusion in a closed loop would allow direct optimization of designed flexibility, building on NMA-tune [26] and FlexFlow [25].

Risks and alternatives. *Risk 1: rare events and transient contacts.* BBFlow’s transient-contact Jaccard (29%) trails AlphaFlow-T (47%), suggesting that rare specific contacts may require evolutionary priors. *Mitigation*: introduce an optional pLM channel, fine-tune on contact-rich trajectories, or augment the loss with a contact-prediction term. *Risk 2: dependence on the input structure.* If x_{eq} is unavailable, a single AlphaFold 2 prediction suffices and the AlphaFold + BBFlow pipeline remains $\sim 30\times$ faster than AlphaFlow at comparable accuracy. Robustness to noisy inputs is verified by adding 0.1–0.2 Å Gaussian noise to backbone coordinates and to using random MD frames as x_{eq} , with only minor degradation. *Risk 3: distributional scope.* As an MD emulator, BBFlow cannot predict alternative folded states far from x_{eq} without retraining on transition-rich trajectories; specialized models such as Str2Str [16] can be used in that regime, accepting reduced MD-emulation fidelity.

Deliverables. (D1) Trained BBFlow weights, training code, and evaluation scripts released under a permissive licence at <https://github.com/graeter-group/bbflow>; (D2) public benchmark on ATLAS, *de novo*, and multi-chain test sets; (D3) a peer-reviewed publication (NeurIPS 2025); (D4) tutorial materials for incorporating BBFlow into protein design pipelines.

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